THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The common difference of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 4 **(b)** -4 **(c)** 8 **(d)** 5

2. In a right triangle with legs 3 and 4, the hypotenuse is: 1

(a) 5 **(b)** 7 **(c)** 1 **(d)** 25

3. The distance of the point $(3, 4)$ from the origin is: 1

(a) 5 **(b)** 25 **(c)** 1 **(d)** 7

4. The 10th term of the AP $3, 7, 11, 15, \ldots$ is: 1

(a) 39 **(b)** 43 **(c)** 40 **(d)** 35

5. The ratio of corresponding sides of two similar triangles is 1:3. Their perimeters are in the ratio 1: 1

(a) 4 **(b)** 3 **(c)** 1 **(d)** 9

6. The midpoint of the join of $(2, 4)$ and $(6, 8)$ is: 1

(a) $(5, 6)$ **(b)** $(8, 12)$ **(c)** $(4, 6)$ **(d)** $(4, 7)$

7. The sum of the first 10 terms of $1, 2, 3, 4, \ldots$ is: 1

(a) 55 **(b)** 66 **(c)** 110 **(d)** 45

8. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$, $DB = 3$, $AE = 4$. Then $EC =$ 1

(a) 6 **(b)** 8 **(c)** 4 **(d)** 3

9. The point $(3, 4)$ lies in the: 1

(a) third quadrant **(b)** fourth quadrant **(c)** first quadrant **(d)** second quadrant

10. Which of the following sequences is an AP? 1

(a) $4, 7, 11, 15, \ldots$ **(b)** $4, 7, 10, 13, \ldots$ **(c)** $4, 7, 13, 22, \ldots$ **(d)** $4, 7, 13, 16, \ldots$

11. The areas of two similar triangles are in the ratio 4:9. The ratio of their corresponding sides is 2: 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 9

12. The coordinates of the point on the x-axis at distance 3 from the origin on the right are: 1

(a) $(3, 0)$ **(b)** $(-3, 0)$ **(c)** $(0, 3)$ **(d)** $(3, 3)$

13. In an AP with $a = 5$ and $d = 3$, the 2nd term is: 1

(a) 11 **(b)** 15 **(c)** 8 **(d)** 2

14. The diagonal of a rectangle with sides 3 and 4 is: 1

(a) 25 **(b)** 7 **(c)** 1 **(d)** 5

15. The distance between the points $(6, 0)$ and $(0, 8)$ is: 1

(a) 100 **(b)** 2 **(c)** 10 **(d)** 14

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Write the first four terms of the AP with $a = 5$ and $d = 3$. 2

17. Find the hypotenuse of a right triangle with legs 6 cm and 8 cm. 2

18. Find the distance between the points $(0,\ 0)$ and $(3,\ 4)$. 2

19. Find the common difference of the AP $6, 11, 16, 21, \ldots$ 2

20. The sides of two similar triangles are in the ratio 2:3. Find the ratio of their areas. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the 12th term of the AP $3, 7, 11, 15, \ldots$ 3

22. In a right triangle the two perpendicular sides are 3 cm and 4 cm. Find the hypotenuse and the perimeter. 3

23. Find the distance between the points $(-2,\ 1)$ and $(1,\ 5)$. 3

24. Find the sum of the first 10 terms of the AP $2, 5, 8, 11, \ldots$ 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, find EC. 3

25. Find the midpoint of the segment joining $(1,\ 7)$ and $(9,\ 3)$. 3

OR

Which term of the AP $4, 9, 14, 19, \ldots$ is 54? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. How many terms are there in the AP $7, 10, 13, \ldots, 64$? 5

OR

A ladder 5 m long reaches a window 4 m above the ground. Find the distance of the foot of the ladder from the wall. 5

27. Find the area of the triangle with vertices $(0,\ 0)$, $(4,\ 0)$ and $(0,\ 3)$. 5

OR

In an AP, $a = 6$, $d = 4$, $n = 20$. Find a_n and S_n. 5

END OF QUESTION PAPER — **Total: 50 Marks**